Title

Brute Force Attack Detection using Splunk SIEM

A cybersecurity detection project focused on identifying brute force login attempts through repeated failed authentication events, suspicious IP behavior, and alert generation.

Key Features

- Detect repeated failed login attempts
- Identify suspicious source IP addresses
- Threshold-based alerting
- SOC analyst investigation use case
- SIEM / log analysis ready

Technologies Used

- Python
- Splunk / SIEM
- Windows Event Logs / Linux Auth Logs
- Log Parsing & Detection Logic

MITRE ATT&CK Mapping

T1110 – Brute Force

Purpose

Built for educational use, blue-team practice, and security monitoring in authorized lab environments.

Author

Chaitanya
Cybersecurity Enthusiast

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Executive Summary

1.1 Project Overview

This project was designed to simulate a real-world brute-force attack scenario in a controlled lab environment and demonstrate how a Security Operations Center (SOC) analyst can detect suspicious authentication activity using Splunk Enterprise. The lab consisted of three main components: an attacker machine running Kali Linux, a Windows 10 virtual machine acting as the target system, and Splunk installed for centralized log collection and analysis.

The attack simulation involved generating multiple failed login attempts against user accounts on the Windows system. These repeated attempts created Windows Security logs, which were then ingested into Splunk for investigation. By using Windows authentication event data, the project aimed to replicate a common threat scenario faced by organizations: password brute-force attacks targeting user credentials.

The primary focus of the project was not only to launch the attack but also to build the complete detection workflow. This included configuring the environment, ensuring network connectivity between systems, enabling Windows log collection, indexing data in Splunk, and performing searches to identify suspicious activity. The project reflects how SOC teams rely on log sources and SIEM tools to detect threats in enterprise environments.

1.2 Key Outcomes

The project successfully achieved its objective of detecting brute-force activity using Windows Security logs. After logs were ingested into Splunk, searches were performed to identify Event ID 4625, which represents failed login attempts on Windows systems. Analysis of the results showed repeated failed attempts against accounts such as testuser, indicating brute-force behavior.

In addition, the source IP address of the attacking system was visible in the logs, allowing attribution of the suspicious activity to the Kali Linux machine used during testing. This demonstrates the importance of centralized logging and visibility in cybersecurity operations.

Through this project, practical skills were developed in areas such as SIEM deployment, Windows log analysis, Splunk query writing, attack detection, and troubleshooting ingestion issues. Overall, the project successfully simulated a realistic SOC use case and provided hands-on experience with threat monitoring and incident detection.

Introduction

2.1 Brute Force Attack Overview

A brute force attack is a common cyberattack technique in which an attacker repeatedly tries multiple username and password combinations in an attempt to gain unauthorized access to a system or account. This method relies on trial and error and is often automated using specialized tools. Brute force attacks usually target login services such as Remote Desktop Protocol (RDP), Secure Shell (SSH), web applications, or Windows user accounts. If weak or predictable passwords are being used, the chances of a successful compromise become much higher.

Brute force attacks are dangerous because they can lead to unauthorized access, privilege escalation, data theft, and further compromise of internal systems. Even when the attack is unsuccessful, repeated failed login attempts generate suspicious activity that should be monitored by security teams. Organizations commonly defend against such attacks by enforcing strong password policies, account lockout policies, multi-factor authentication (MFA), and continuous monitoring of authentication logs.

In this project, a brute force attack was simulated in a controlled lab environment against a Windows machine. The objective was not to gain unauthorized access, but to generate failed login events that could later be detected and investigated using Splunk Enterprise.

2.2 Project Objective

The primary objective of this project was to build a practical SOC lab that demonstrates how brute force attacks can be detected using log analysis and SIEM tools. A virtual environment was created using Kali Linux as the attacker machine and Windows 10 as the victim machine. Splunk Enterprise was used to collect, index, and analyze security logs generated during the attack simulation.

The project focused on understanding how Windows authentication logs record failed login attempts, particularly Windows Security Event ID 4625. By collecting these logs in Splunk, it became possible to identify targeted user accounts, count repeated failed attempts, and trace the source IP address of the attacker system.

Another objective was to gain hands-on experience with core SOC analyst tasks such as log ingestion, event monitoring, writing SPL queries, and identifying indicators of suspicious behavior. This project demonstrates a real-world workflow where attacks are detected through centralized logging and investigation, making it highly relevant for cybersecurity internships and SOC analyst roles.

Lab Setup

3.1 Environment Architecture

This project was developed in a controlled virtual lab environment to simulate a real-world Security Operations Center (SOC) scenario. The lab consisted of multiple virtual machines connected through the same internal network. A separate attacker machine was used to generate malicious activity, while a Windows machine acted as the target system. Splunk Enterprise was used as the Security Information and Event Management (SIEM) solution to collect, monitor, and analyze system logs. This architecture helped create a safe environment for learning cybersecurity concepts such as attack detection, log monitoring, and incident investigation without affecting any real production systems.

3.2 Tools Used

Several tools were used to complete this project successfully. Kali Linux was used as the attacker machine because it contains many built-in cybersecurity tools for testing and simulation. CrackMapExec was used to simulate brute-force login attempts against the Windows target system. Windows 10 was used as the victim machine where authentication logs were generated. Splunk Enterprise was used to ingest and analyze Windows event logs, especially Security logs related to failed login attempts. VirtualBox was used to host and manage all virtual machines in a single system. Together, these tools created a complete SOC home lab environment for practical learning.

3.3 Network Configuration

Proper network configuration was an important part of the project. Both the Kali Linux and Windows virtual machines were connected to the same virtual network using the virtualization platform settings. This allowed both systems to communicate with each other as if they were connected in a real local area network. Connectivity was tested using the ping command to verify successful communication between machines. Once network access was confirmed, the attacker machine could reach the Windows target and perform authentication attempts. The same network connectivity also allowed logs and data to be accessed and monitored through Splunk. Correct network setup ensured smooth attack simulation, accurate log generation, and effective threat detection throughout the project.

Methodology

4.1 Splunk Installation & Configuration

The first step of the project was installing Splunk Enterprise on the Windows virtual machine. Splunk was selected as the SIEM platform because it is widely used for log monitoring, security analysis, and threat detection. After installation, the Splunk service was started and accessed through the web interface on port 8000. Initial verification was performed to confirm that the Splunk daemon and web services were running properly. Basic configuration included creating a local admin account, accessing the Search & Reporting application, and confirming that internal Splunk logs were being indexed. This ensured that the platform was operational before adding Windows security logs for analysis.

4.2 User Account Creation

To simulate a realistic brute-force attack scenario, test user accounts were created on the Windows machine. These accounts acted as attack targets during the password guessing phase. A sample account such as testuser was created using Windows command-line tools. Creating dedicated test accounts helped avoid impacting the main administrator account while generating authentication logs safely. These accounts also allowed multiple login attempts, enabling the collection of repeated failed login events. Using separate user accounts is a common practice in lab environments because it creates realistic attack conditions while maintaining system stability and administrative control.

4.3 Log Ingestion Setup

The next step was configuring Windows Event Logs to be ingested into Splunk. Since some user interface methods were unavailable, manual configuration was completed using the inputs.conf file located in the Splunk configuration directory. Windows Security, System, and Application logs were enabled for collection. Security logs were the most important source because they contain authentication events such as successful and failed logins. After updating the configuration file, Splunk was restarted so that the new settings could take effect. Once restarted, thousands of events were visible in the Splunk index, confirming successful data ingestion from the Windows host into the SIEM platform.

4.4 Attack Simulation

After log ingestion was confirmed, a brute-force attack was simulated from Kali Linux against the Windows machine. Multiple password attempts were made against the created user accounts using a controlled password list. This generated failed login events inside Windows Security logs. Because the attack was performed in a lab environment, it provided a safe and legal way to understand how authentication attacks appear in SIEM tools.

Log Analysis & Detection

5.1 Windows Event ID 4625

Windows Security Event ID 4625 represents a failed login attempt. This event is generated whenever a user enters incorrect credentials or authentication fails for another reason. It is one of the most important event IDs for SOC analysts because repeated 4625 events often indicate password guessing, brute-force attacks, or unauthorized access attempts. During this project, Event ID 4625 was the primary indicator used to detect suspicious login activity generated during the simulated attack.

5.2 Detection Query Used

To identify brute-force behavior, a Splunk Search Processing Language (SPL) query was used to filter failed login events and count repeated attempts. The query grouped logs by targeted account name and source IP address. This made it easier to determine which accounts were attacked and from where the attempts originated.

```
index=main EventCode=4625
| stats count by Account_Name, Source_Network_Address
| sort - count
```

This query sorted results in descending order so the most attacked accounts appeared first.

5.3 Identifying Source IP & Target Accounts

The Account_Name field showed the usernames that received failed login attempts, while the Source_Network_Address field identified the IP address of the attacking machine. In this lab, the targeted accounts included the test user accounts created on the Windows machine. The source IP matched the Kali Linux system used to simulate the brute-force activity. This demonstrates how SOC analysts trace suspicious authentication attempts back to their source.

5.4 Event Investigation Process

After detecting repeated failed logins, the next step is investigation. Analysts review the number of attempts, time period, affected accounts, and whether any successful login followed the failures. If the same source IP targets multiple accounts in a short period, it strongly suggests automated brute-force behavior. In a real organization, analysts would also check firewall logs, endpoint alerts, and account lockout events for further evidence.

5.5 Security Significance

Failed login monitoring is a critical detection use case in any SOC environment. Brute-force attempts can lead to unauthorized access if weak passwords are used. By monitoring Event ID 4625 and creating thresholds or alerts, organizations can quickly detect and respond to such threats. This project demonstrated how Splunk can be used to turn raw Windows logs into meaningful security alerts and actionable findings.

Findings & Results

6.1 Failed Login Attempts Detected

The project successfully identified multiple failed login attempts from the Windows Security logs collected in Splunk Enterprise. These events were detected using Windows Event ID 4625, which represents unsuccessful authentication attempts. By searching and filtering these logs, it was confirmed that repeated login failures had occurred during the brute-force simulation. This demonstrated that Splunk was properly ingesting logs and that suspicious authentication activity could be monitored effectively.

6.2 Most Targeted Accounts

Analysis of the logs showed that certain user accounts were targeted more frequently than others. The account named **testuser** recorded the highest number of failed login attempts, while another user account also showed multiple authentication failures. This pattern indicates that attackers often focus on specific accounts that may have weak or predictable credentials. Identifying the most targeted accounts is important because it helps security teams prioritize account monitoring, enforce password resets, and strengthen access controls.

6.3 Attack Analysis

The repeated failed login attempts strongly suggested automated brute-force behavior rather than normal user error. A high number of attempts within a short time frame is commonly associated with password guessing tools or scripted attacks. The source IP address field in the logs helped identify the attacking system, which in this lab environment was the Kali Linux machine. In a real-world SOC environment, such activity would trigger alerts for suspicious authentication behavior. Overall, the results confirmed that Splunk can be used effectively to detect brute-force attacks, investigate affected accounts, and support incident response activities.

Challenges Faced

7.1 Splunk Configuration Issues

During the project, one of the main challenges was configuring Splunk Enterprise properly inside the virtual machine environment. Although the Splunk backend service was running, the web interface initially failed to load correctly due to service startup issues and browser routing errors. Some internal settings pages returned “Page Not Found” messages, which created confusion during the configuration process. This highlighted the importance of understanding the difference between Splunk background services (splunkd) and the web interface (splunkweb). Restarting services, verifying running status, and using the correct access URL helped resolve these issues.

Another challenge involved distinguishing between Splunk Universal Forwarder and the full Splunk Enterprise installation. Since both products use similar command-line tools, commands were initially executed in the wrong installation directory. This caused misleading outputs and delayed troubleshooting. Once the correct Splunk Enterprise path was used, management and restart operations worked properly.

7.2 Log Ingestion Troubleshooting

The second major challenge was ingesting Windows Event Logs into Splunk. Initial attempts through the Splunk web interface using local event log collection pages did not work correctly because of broken or unsupported routes in the installed version. Command-line methods such as splunk add input also failed due to version limitations. As a result, logs were not appearing in searches despite Splunk itself functioning normally.

To solve this issue, a manual configuration method was used by creating an inputs.conf file inside the Splunk configuration directory. Windows Security, System, and Application logs were enabled through this file, and Splunk was restarted. After this change, thousands of events were successfully indexed, confirming that log ingestion was working.

This troubleshooting process was an important learning experience because it demonstrated how SOC analysts often need to solve configuration problems before they can begin actual threat detection. It also reinforced the importance of persistence, methodical testing, and understanding multiple ingestion methods when working with SIEM platforms.

MITRE ATT&CK Mapping

8.1 T1110 – Brute Force

The activities performed during this project align with the MITRE ATT&CK framework under technique **T1110 – Brute Force**. MITRE ATT&CK is a globally recognized knowledge base used by cybersecurity professionals to classify attacker behaviors, tactics, and techniques observed during real-world intrusions. Mapping security events to this framework helps SOC analysts understand attacker intent, improve detections, and communicate incidents in a standardized format.

Brute Force (T1110) refers to attempts by an attacker to gain unauthorized access by systematically trying multiple passwords or authentication combinations against one or more accounts. These attacks may target local accounts, domain accounts, VPN portals, Remote Desktop services, SSH, SMB shares, web applications, or any login interface that accepts credentials. Attackers often automate these attempts using scripts or offensive tools to test large password lists quickly.

In this project, repeated login attempts were generated from the attacker machine against user accounts hosted on the Windows target system. These attempts resulted in multiple failed authentication events, which were recorded in Windows Security Logs and forwarded into Splunk Enterprise. The key evidence of this behavior was **Windows Event ID 4625**, which indicates failed logon attempts.

The following Splunk query was used to detect this activity:

```
index=main EventCode=4625
| stats count by Account_Name, Source_Network_Address
| sort - count
```

This query helped identify the most targeted accounts and the IP address responsible for the login failures. High volumes of failed attempts against specific users are common indicators of brute-force activity.

From a SOC perspective, mapping this detection to T1110 is important because it allows defenders to categorize the incident correctly and respond with suitable actions such as account lockout policies, password resets, MFA enforcement, firewall blocks, and alert tuning.

Overall, this project demonstrated how brute-force attacks can be simulated in a controlled lab and accurately mapped to MITRE ATT&CK T1110 using log analysis in Splunk.

Conclusion

9.1 Project Summary

This project successfully demonstrated how a Security Operations Center (SOC) workflow can be implemented in a home lab environment using Kali Linux, Windows 10, and Splunk Enterprise. The main objective was to simulate a brute-force login attack and detect the activity using centralized log monitoring. A controlled attack was launched from the Kali Linux machine against a Windows user account, generating multiple failed authentication attempts. These events were then collected and analyzed in Splunk using Windows Security logs.

The project provided practical experience in configuring a SIEM platform, ingesting Windows event logs, and writing detection queries. By using Event ID 4625, repeated failed login attempts were successfully identified. The analysis also helped determine which user accounts were targeted most frequently and from which source IP address the attempts originated. This mirrors real-world SOC analyst responsibilities such as monitoring authentication activity, investigating suspicious behavior, and identifying potential brute-force attacks before successful compromise occurs.

In addition to attack detection, the project also improved troubleshooting skills. Several challenges related to Splunk configuration, data inputs, and log ingestion were resolved during implementation. This highlighted the importance of persistence, problem-solving, and understanding log pipelines in security operations. Overall, the project delivered a complete hands-on experience covering attack simulation, log collection, threat detection, and incident analysis.

9.2 Future Improvements

This project can be enhanced further by implementing real-time alerts in Splunk Enterprise whenever failed login attempts exceed a defined threshold. Dashboards can also be created to visualize trends such as top attacked accounts, source IP addresses, and login failures over time. Additional attack scenarios such as privilege escalation, malware execution, or suspicious PowerShell activity can be added to expand the SOC lab. Integrating other tools such as Sysmon or Wireshark would provide deeper visibility into host and network events. These improvements would make the project more advanced and closer to a real enterprise SOC environment.

Appendix

10.1 Queries Used

The appendix includes all Splunk Processing Language (SPL) queries used during the project for log verification, brute-force detection, and analysis. These queries helped identify failed login attempts, targeted accounts, and source IP addresses. The primary detection query used was:

```
index=main EventCode=4625
| stats count by Account_Name, Source_Network_Address
| sort - count
```

This query filters Windows Security logs for Event ID 4625, which represents failed login attempts, and groups the results by username and source IP address. Additional queries such as index=main were used to verify successful log ingestion, while index=main EventCode=4624 can be used to review successful logins. Including these queries in the appendix allows readers to understand the analytical methods used during the investigation.

10.2 Screenshots

Relevant screenshots should be attached in this section to provide visual evidence of project execution and results. Screenshots may include the Splunk dashboard, indexed events, query results showing failed login attempts, and the virtual lab environment setup. A screenshot of the detection query output is especially useful because it demonstrates that the brute-force attempts were successfully detected. Additional screenshots may include the Windows VM, Kali Linux terminal, and Splunk configuration files used for log ingestion.

Screenshots make the report more professional and easier to validate. They also help interviewers, recruiters, or reviewers quickly understand the practical work completed in the project. Each screenshot should be labeled clearly with a short title such as “Splunk Search Results,” “Failed Login Detection,” or “Lab Architecture.”

10.3 References

This project used publicly available documentation and frameworks for learning and implementation. Key references include official Splunk Enterprise documentation for installation, configuration, and SPL query syntax. Microsoft Windows Event documentation was used to understand Security Event IDs such as 4625 (failed login) and 4624 (successful login). The MITRE ATT&CK framework was referenced to map the activity to technique T1110 (Brute Force).

Other useful references may include tutorials on Windows logging, SIEM best practices, and cybersecurity home lab design. Including references increases the credibility of the report and shows that standard industry resources were used during project development.

```
┌──(chaitanya㉿kali)-[~]
└─$ crackmapexec smb 192.168.1.7 -u testuser -p /home/chaitanya/Desktop/file.txt
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [*] Windows 10 / Server 2019 Build 19041 x64 (name:DESKTOP-A5H24G4) (domain:DESKTOP-A5H24G4) (signing:False) (SMBv1:False)
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [-] DESKTOP-A5H24G4\testuser:123456 STATUS_LOGON_FAILURE
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [-] DESKTOP-A5H24G4\testuser:123 STATUS_LOGON_FAILURE
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [-] DESKTOP-A5H24G4\testuser:24586976 STATUS_LOGON_FAILURE
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [-] DESKTOP-A5H24G4\testuser:16154684864 STATUS_LOGON_FAILURE
SMB         192.168.1.7     445    DESKTOP-A5H24G4  [+] DESKTOP-A5H24G4\testuser:password@123

┌──(chaitanya㉿kali)-[~]
└─$
```

Data inputs - Settings | Splunk

localhost:8000/en-US/manager/search/datainputstats

splunk>enterprise Apps Administra... 1 Messages Settings Activity Help Find

Data inputs

Set up data inputs from files and directories, network ports, and scripted inputs. If you want to set up forwarding and receiving between two Splunk instances, go to Forwarding and receiving.

Local inputs

Type	Inputs	Actions
Local event log collection Collect event logs from this machine.	-	Edit
Remote event log collections Collect event logs from remote hosts. Note: this uses WMI and requires a domain account.	1	+ Add new
Files & Directories Index a local file or monitor an entire directory.	19	+ Add new
Local performance monitoring Collect performance data from local machine.	0	+ Add new
Remote performance monitoring Collect performance and event information from remote hosts. Requires domain credentials.	0	+ Add new
HTTP Event Collector	0	+ Add new

localhost:8000/en-US/manager/search/adddata/selectsource?input_type=perfmon_remote&input_mode=1

Type here to search Gru... ENG 15:09 25-04-202